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(54) Title: PROCESS FOR THE PREPARATION OF CENOBAMATE AND INTERMEDIATES THEREOF

(57) Abstract: The present invention relates to a process for the preparation of Cenobamate and intermediates thereof.

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**“PROCESS FOR THE PREPARATION OF CENOBAMATE AND
INTERMEDIATES THEREOF”**

Related applications:

This application claims the benefit under Indian Provisional Application No's.
5 202241007524, filed on Feb 12, 2022 and 202241027288, filed on May 12, 2022, the
contents of which are incorporated by reference herein.

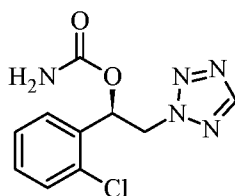
Field of the invention:

The present invention relates to a process for the preparation of Cenobamate and
10 intermediates thereof.

Background of the invention:

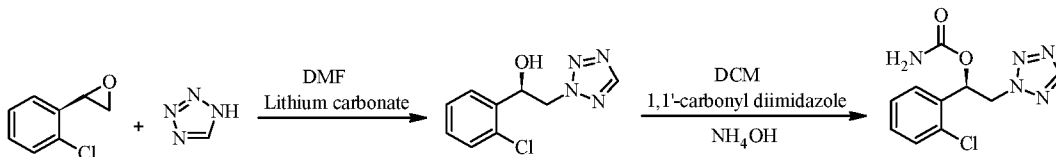
Cenobamate is a novel new antiepileptic drug, which can be used in the treatment of
disorders of the central nervous system, especially as anxiety, depression, convulsion,
15 epilepsy, migraine, bipolar disorder, drug abuse, smoking, ADHD, obesity, sleep disorder,
neuropathic pain, stroke, cognitive impairment, neurodegeneration, stroke and muscle
spasm.

Cenobamate is chemically known as [(1R)-1-(2-Chlorophenyl)-2-(tetrazol-2-yl) ethyl]
20 carbamate and is structurally represented by the following formula I.



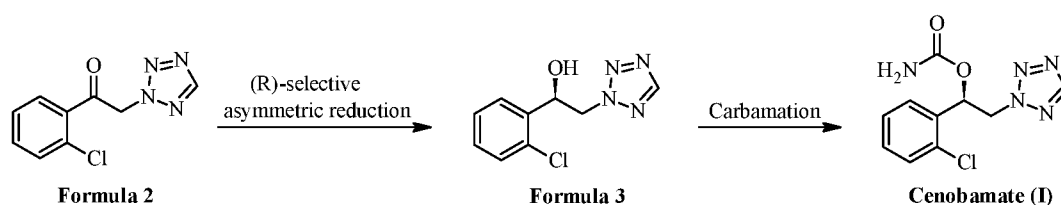
Formula I

Cenobamate was approved in US under the brand name Xcopri. Cenobamate and process
for its preparation was first disclosed in US 7598279 (the US '279 patent) and the disclosed
25 process is schematically depicted in the following scheme:



The process reported in US '945 involves reaction of (R)-2-chlorostyrene oxide with 1H-tetrazole in lithium carbonate provides (R)-configured alcohol compound, which on reaction with 1,1'-carbonyl diimidazole in methylene chloride yields Cenobamate. The said process involves column chromatography purification for the preparation of Cenobamate, and not be suitable for large scale production.

The US 8501436 patent (the US '436 patent) discloses a preparation method of carbamic acid (R)-1-aryl-2-tetrazolyl-ethyl ester (I), comprising the asymmetric reduction of arylketone (2), followed by carbamation of alcohol (3); the disclosed process is schematically depicted in the following scheme:



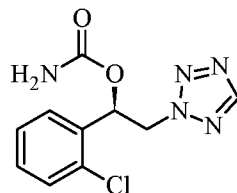
However, there is always a need for an alternative process, which for example, involves use of reagents/solvents that are less expensive and easier to handle, consume smaller amounts of solvents, and provide a higher yield of product with higher purity. Hence, the main objective of the present invention is to provide cost effective and commercially viable process for the preparation of Cenobamate of Formula I.

Summary of the invention:

Accordingly, the present invention provides a novel process for the preparation of Cenobamate of formula 1, which is cost effective and commercially viable.

In one aspect, the present invention provides a process for the preparation of Cenobamate of formula I, which comprises:

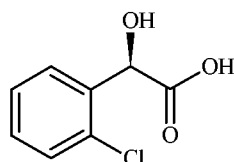
- reacting (R)-(-)-2-chloromandelic acid of formula II with an alcohol in the presence of a catalyst to provide a compound of formula III;
- optionally, reacting the compound of formula III with a suitable hydroxy protecting agent to provide a compound of formula IV;
- reducing the compound of formula IV or the compound of formula III with a suitable reducing agent to provide a compound of formula V;



Formula I

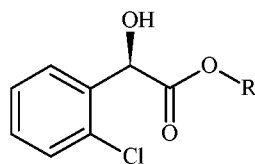
which comprises:

- a) reacting (R)-(-)-2-Chloromandelic acid of formula II,



Formula II

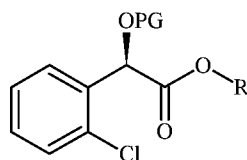
with a suitable alcohol in the presence of a catalyst to provide a compound of formula III,



Formula III

wherein R is C₁ to C₆ alkyl,

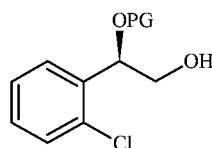
- b) optionally, reacting the compound of formula III with a suitable hydroxy protecting agent to provide a compound of formula IV,



Formula IV

wherein R is C₁ to C₆ alkyl and PG is a hydroxy protecting group,

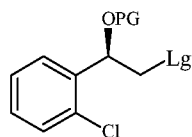
- c) reducing the compound of formula IV or the compound of formula III with a suitable reducing agent to provide a compound of formula V,



Formula V

wherein PG is hydrogen or hydroxy protecting group,

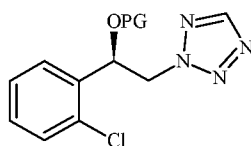
- d) reacting the compound of formula V with a suitable activating agent to provide a compound of formula VI,



Formula VI

wherein PG is hydrogen or hydroxy protecting group and Lg is leaving group,

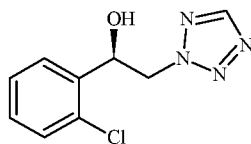
- e) reacting the compound of formula VI with tetrazole or its salt to provide a compound of formula VII,



Formula VII

wherein PG is hydrogen or hydroxy protecting group,

- f) deprotecting the compound of formula VII, if PG is hydroxy protecting group to provide a compound of formula VIII; and



Formula VIII

- g) converting the compound of formula VIII into Cenobamate of formula I.

The starting (R)-(-)-2-Chloromandelic acid of formula II is known in the art and is commercially available. The same can be obtained from available commercial sources.

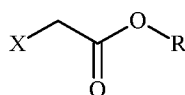
- 20 The step a) of aforementioned process involves the reaction of (R)-(-)-2-chloromandelic acid of formula II with a suitable alcohol selected from but not limited to methanol, ethanol, propanol and the like; preferably methanol; in the presence of a catalyst selected from acids such as hydrochloric acid, sulphuric acid, phosphoric acid and the like; preferably sulphuric acid; under appropriate reaction conditions to provide a compound of
- 25 formula III, wherein R is alkyl selected from methyl, ethyl, propyl and the like; preferably methyl. The said reaction is carried out at a suitable temperature of about 25°C to about

In another embodiment, the present invention provides crystalline form of Cenobamate characterized by the PXRD pattern shown in Fig-1, herein designated as crystalline form M. The crystalline form M of Cenobamate further characterized by a powder X-ray diffractogram that comprises one or more peaks at 10.03, 12.37, 13.44, 14.73, 17.07, 18.34, 20.27, 21.84, 23.03 24.11, 26.47 and 27.03 degrees $2\theta \pm 0.2$.

In an embodiment, the Cenobamate obtained by the present invention having chemical purity greater than 99% by HPLC; and having optical purity of greater than 99.9% by HPLC.

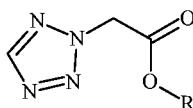
In another embodiment, the present invention provides a process for the preparation of Cenobamate of formula I, which comprises:

a) reacting tetrazole with alkylhaloacetate of formula IX, wherein X is halogen; R is alkyl,



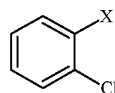
Formula IX

in the presence of base in a suitable solvent to provide a compound of formula X, wherein R is alkyl;



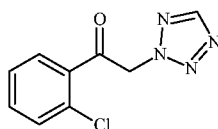
Formula X

b) reacting the compound of formula X with 1-halo-2-chlorobenzene of formula XI, wherein X is halogen;



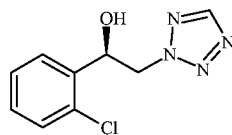
Formula XI

in the presence of a suitable organolithium reagent in a suitable solvent to provide a compound of formula XII; and



Formula XII

c) converting the compound of formula XII into compound of formula VIII, and



Formula VIII

d) converting the compound of formula VIII into Cenobamate of formula I.

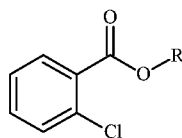
- 5 The step a) of aforementioned process involves the reaction of tetrazole with alkylhaloacetate of formula IX, wherein X is halogen; R is alkyl; in the presence of a suitable base in a suitable solvent under appropriate conditions to provide a compound of formula X, wherein R is alkyl having 1 to 6 carbon atoms.

Further, 1N positional isomer of formula X is also formed along with formula X and the
10 same is separated and removed by way of recrystallisation or other available purifications methods such as purified using silica gel chromatography. Preferably purified using silica gel chromatography (Gradient: ethyl acetate in hexane) before proceeding to the next step.

The step b) of aforementioned process involves the reaction of compound of formula X with 1-halo-2-chlorobenzene of formula XI, wherein X is halogen; in the presence of a
15 suitable organolithium reagent selected from but not limited to $i\text{PrMgCl} \cdot \text{LiCl}$, n-butyl lithium, sec-butyllithium, isopropyllithium, tert-butyllithium, phenyl lithium and the like; in a suitable solvent under appropriate conditions to provide a compound of formula XII.

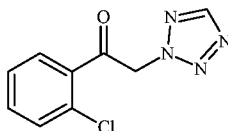
The step c) of aforementioned process involves the conversion of compound of formula XII into compound of formula VIII. The said conversion can be carried out by the methods
20 known in the art such as subjecting an arylketone of formula XII to (R)-selective asymmetric reduction is achieved by biological asymmetric reduction or by chemical asymmetric reduction with a chiral borane reductant, or by asymmetric catalytic hydrogenation, or by asymmetric catalytic transfer hydrogenation to form an alcohol of formula VIII.

25 The step d) of aforementioned process involves the conversion of compound of formula VIII into cenobamate of compound formula I can be carryout by the methods known in the art.



Formula XIX

- b) reacting the compound of formula XIX with Lithium (trimethylsilyl)diazomethane to provide compound of formula XII, and



Formula XII

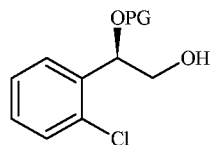
- c) converting the compound of formula XII into cenobamate of formula I.

The step a) of aforementioned process involves the reaction of compound of formula XVIII in the presence of a suitable alcohol selected from but not limited to methanol, ethanol, propanol and the like; in the presence of a catalyst selected from acids such as hydrochloric acid, sulphuric acid, phosphoric acid and the like under appropriate reaction conditions to provide a compound of formula XIX, wherein R is alkyl selected from methyl, ethyl, propyl and the like.

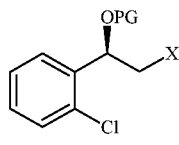
The step b) of aforementioned process involves the reaction of compound of formula XIX with Lithium (trimethylsilyl)diazomethane to provide a compound of formula XII. The reaction can be carried out in presence or absence of a base in a suitable solvent under appropriate reaction conditions. The said reaction directly provides the required compound of formula XII without formation additional 1N-positional isomer and thereby avoiding tedious column purifications/recrystallisation in order to isolate and separate the 1N position isomer from the required 2N positional isomer compound as in the prior art.

The step c) of aforementioned process involves conversion of compound of formula XII into cenobamate of formula I as described in step c) & step d) of the second embodiment of the present invention or by the methods known in the art.

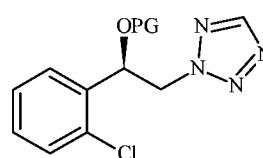
In a further embodiment, the present invention provides compounds of formula V, formula VI and formula VII represented by the following structural formula, which are useful in the preparation of Cenobamate of formula I.



Formula V;



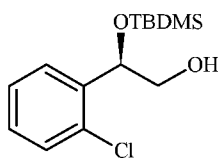
Formula VI;



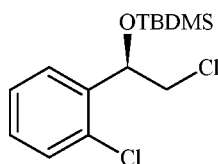
Formula VII;

wherein PG is a hydroxy protecting group and X is halogen

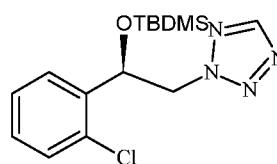
In a preferred embodiment, the present invention provides compounds of formula Vb, formula VIb and formula VIIb represented by the following structural formula, which are useful in the preparation of Cenobamate of formula I.



Formula Vb



Formula VIb



Formula VIIb

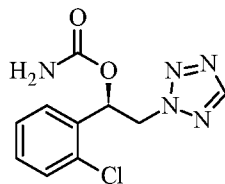
5 wherein TBDMS is tert-butyldimethylsilyl group.

In a further embodiment, the present invention provides use of compounds of formula V, formula VI, formula VII, formula Vb, formula VIb and formula VIIb as an intermediate for the preparation of Cenobamate of formula I.

- 10 In another embodiment, the present invention provides a process for the preparation of (2R)-2-(2-chlorophenyl)oxirane, which comprises of the following steps;
- a) esterification of (R)-(-)-2-Chloromandelic acid with a suitable alcohol such as methanol, ethanol, isopropanol and the like; in the presence of a suitable acid catalyst such as hydrochloric acid, sulphuric acid and the like; to provide a corresponding alkyl
15 (2R)-(2-chlorophenyl)(hydroxy)acetate;
 - b) reducing the alkyl (2R)-(2-chlorophenyl)(hydroxy)acetate with a suitable reducing agent such as sodium borohydride, sodiumcyanoborohydride and the like in a suitable solvent; to provide a (1R)-1-(2-chlorophenyl)ethane-1,2-diol; and
 - c) reacting the (1R)-1-(2-chlorophenyl)ethane-1,2-diol with trimethylorthoformate and
20 trimethylsilylchloride in a suitable solvent followed by treating with a suitable base to provide (2R)-2-(2-chlorophenyl)oxirane.

WE CLAIM:

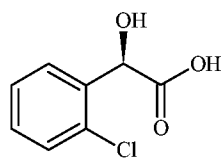
1. A process for the preparation of Cenobamate of formula I,



Formula I

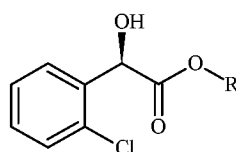
5 which comprises:

- a) reacting (R)-(-)-2-Chloromandelic acid of formula II,



Formula II

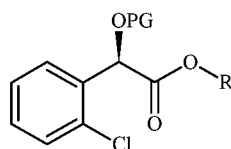
with a suitable alcohol to provide a compound of formula III,



Formula III

wherein R is C₁ to C₆ alkyl,

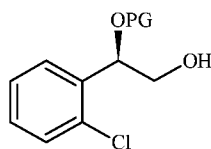
- b) optionally, reacting the compound of formula III with a suitable hydroxy protecting agent to provide a compound of formula IV,



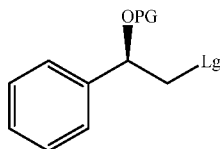
Formula IV

wherein R is C₁ to C₆ alkyl and PG is a hydroxy protecting group,

- c) reducing the compound of formula IV or the compound of formula III with a suitable reducing agent to provide a compound of formula V,

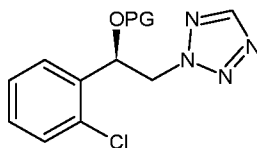


Formula V



Formula VI

with tetrazole or a salt thereof, in the presence of a base and a solvent to provide a compound of formula VII,



Formula VII

wherein PG is hydrogen or hydroxy protecting group.

12. The process as claimed in claim 11, wherein the leaving group is selected from chloride, bromide, iodide, fluoride, mesylate, tosylate, 4-nitrophenyl sulfonate and trifluoromethyl sulfonate.

13. The process as claimed in claim 11, wherein the base is selected from diisopropylethylamine or potassium tertiarybutoxide and the solvent is N-methyl-2-pyrrolidone.

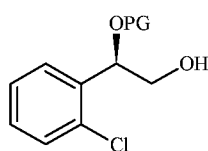
14. A process for the purification of Cenobamate of formula I, which comprises

- a) dissolving Cenobamate of formula I in a suitable solvent;
- b) adding an anti-solvent to the resulting solution; and
- c) isolating the pure Cenobamate of formula I.

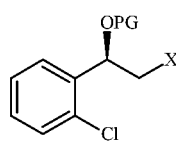
15. The process as claimed in claim 14, wherein the suitable solvent in step a) is selected from alcohols such as methanol, ethanol, isopropanol and butanol; and the anti-solvent is selected from hydrocarbons such as hexane, heptanes, cyclohexane and mixtures thereof.

16. A crystalline form M of Cenobamate characterized by a powder X-ray diffractogram substantially as shown in Figure 1.

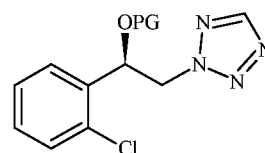
17. A compound of formulae selected from:



Formula V



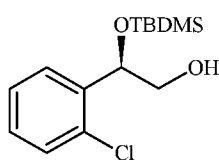
Formula VI



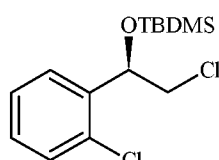
Formula VII

wherein PG is a hydroxy protecting group and X is a leaving group.

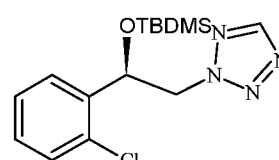
18. The compound as claimed in claim 17 are selected from:



Formula Vb



Formula VIb



Formula VIIb

5 wherein TBDMS is tert-butyldimethylsilyl group.

19. The use compounds as claimed in claim 17 or claim 18 as an intermediate for the preparation of Cenobamate of Formula I.